SITHUM MANUSHA

SOFTWARE ENGINEERING INTERN

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ABOUT ME

Third-year undergraduate studying Information Technology and Management at the University of Moratuwa, passionate about robust software architecture and full-stack development. Experienced in engineering reliable frontend interfaces and backend services using React, Next.js, Java Spring Boot, Node.js, and modern PHP (MVC). Seeking a Software Engineering Internship where I can contribute to real-world production systems and grow within an engineering team.

EDUCATION

University of Moratuwa (UOM) - Moratuwa, Sri Lanka **2024 - Present**
B.Sc. (Hons) in Information Technology & Management **(3rd Year | CGPA: 3.23 / 4.0)**

Open University of Sri Lanka (OUSL) - Nawala, Sri Lanka **2025 - Present**
Bachelor of Laws Honours Degree Programme (LL.B.)

St. Joseph's Vaz College - Wennappuwa, Sri Lanka **2009 - 2022**
G.C.E. Advanced Level (2 A's & 1 B) | G.C.E. Ordinary Level (8 A's & 1 B)

FEATURED PROJECTS

1. Super POS — Multi-Branch Enterprise Offline-First Retail System (Group Project) **2026**

- Built a client-side offline sync queue (**useOfflineSync** with **IndexedDB offlineQueue**) and batch retry pipelines (**/sync/push, /sync/retry**) to buffer offline sales and inventory updates with auto-reconnection sync to **PostgreSQL**.
- Implemented **Last-Write-Wins (LWW)** conflict detection with an interactive 3-way conflict resolver (**Trust Local, Trust Cloud, and Smart Merge**) and **Recharts** dashboards tracking sync throughput and latency.
- Developed a live node status monitor and fleet broadcast system via **Socket.io WebSockets**, alongside device access revocation, entity sync policies, and audit logging with **PDF health report exports**.
- **Technologies:** Next.js, React, TypeScript, Tailwind CSS, IndexedDB, Recharts, NestJS, Node.js, Prisma ORM, Socket.io, Zod, PostgreSQL, Turborepo
- **Live Demo:** pos-system-pos-web-app.vercel.app | **GitHub:** github.com/Ryzera/Ryzera-POS

2. CareerFlow — Full-Stack ATS Resume Optimizer & AI Interview Platform (Individual Project) **2026**

- Built a full-stack platform with a **Java 21 Spring Boot 3** REST gateway, **Python 3.12 FastAPI** NLP worker, and **Next.js 14** client, containerized with **Docker Compose** and automated with **GitHub Actions CI/CD**.
- Developed an automated ATS resume scanner using **TF-IDF vectorization & Cosine Similarity (Scikit-learn)** to detect missing keywords, score job alignment, and evaluate format health.
- Created an interactive AI mock interview simulator with API rate limiting and **26 automated unit and integration tests (JUnit 5, Pytest, Playwright)**.
- **Technologies:** Java 21, Spring Boot 3, Python 3.12, FastAPI, Next.js 14, PostgreSQL, Docker Compose, GitHub Actions, Scikit-learn, JUnit 5, Pytest, Playwright
- **Live Demo:** careerflow-pink.vercel.app | **GitHub:** github.com/SithumManusha/careerflow

3. Ledgerly — Personal Finance & Shared Expense Platform (Individual Project) **2026**

- Built a personal finance tracking web application that monitors monthly expenses, calculates category budget health scores, and alerts users when spending limits are reached.
- Developed an interactive balance projector using **Recharts** to visualize 6-month savings trends based on customizable monthly contributions and expense adjustments.
- Implemented group expense splitting across 4 distribution methods with automated PDF statement exports, backed by a type-safe stack (**React, TypeScript, Node.js, Express, PostgreSQL**) and **60+ Vitest tests**.
- **Technologies:** React, TypeScript, Node.js, Express, PostgreSQL, Drizzle ORM, Recharts, Vitest, Render
- **Live Demo:** ledgerly-nine-rust.vercel.app | **GitHub:** github.com/SithumManusha/Ledgerly

4. Pulse AI Blog Engine — AI Content Platform & MVC Architecture (Individual Project) **2026**

- Architected a full-stack content publishing platform using pure PHP 8 **MVC architecture** (Front Controller routing, Controller-Service separation, dynamic view rendering, and session management).
- Integrated **Google Gemini 1.5 Flash API** via cURL for automated post drafting, 1-click TL;DR bullet summaries, and tag extraction, alongside browser-native **Web Speech API** text-to-speech audio playback.
- Hardened security using **PDO prepared statements** against SQL injection, session-based **CSRF protection**, Markdown **XSS sanitization**, and automated **GitHub CodeQL** vulnerability scanning.
- **Technologies:** PHP 8, MySQL, SQLite (PDO), Google Gemini AI, Docker, GitHub Actions, CodeQL, Tailwind CSS, Web Speech API
- **Live Demo:** pulse-ai-blog-engine.onrender.com | **GitHub:** github.com/SithumManusha/pulse-ai-blog-engine

5. Automated Robotic Sorting System — Microcontroller Conveyor Platform (Group Project) **2025**

Dual-Microcontroller Mechatronics & RFID Distribution

- Constructed an automated conveyor sorting prototype equipped with a 4-DOF robotic arm, ultrasonic sensors, an RFID reader, and 3 sorting compartments.
- Engineered a two-finger mechanical gripper driven by an Arduino and stepper motor, calibrating motor step counts to achieve 98% pick-and-place precision across 3 RFID object categories.
- Tested sensor signal timings and synchronized robotic arm pick-and-place routines with continuous conveyor belt movement.
- **Technologies:** Arduino Mega/Uno (C++), Stepper Motors (NEMA 17, 28BYJ-48), ULN2003, Servo Motors, Integration Testing
- **GitHub:** github.com/SithumManusha/robotic-sorting-system

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, PHP, C, SQL
- **Front-end Development:** React, Next.js 14, TypeScript, Tailwind CSS, Recharts, IndexedDB, State Management
- **Back-end Development:** Spring Boot 3, FastAPI, Node.js, Express.js, NestJS, PHP (OOP / MVC), Socket.io, RESTful APIs
- **Databases & ORM:** PostgreSQL (Supabase), MySQL, SQLite, Spring Data JPA, Hibernate, Prisma ORM, Drizzle ORM, PDO
- **Testing & Quality:** JUnit 5, Pytest, Playwright (E2E), Vitest, Postman API Testing, CodeQL Security Scanning
- **DevOps & Tools:** Docker, Docker Compose, Git, GitHub Actions (CI/CD), Turborepo, Swagger / OpenAPI, Render

CORE COMPETENCIES & ARCHITECTURE

MVC & Microservices Architecture, RESTful API Design, Agile / Scrum, Object-Oriented Design (SOLID), Clean Code, Automated Testing, Offline-First Sync

CERTIFICATIONS

- AWS Cloud Practitioner Essentials – Amazon Web Services (AWS)
- Software Testing with Generative AI – Simplilearn
- Introduction to Artificial Intelligence – Simplilearn
- Introduction to Cybersecurity – Cisco Networking Academy

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Member of the Organizing Committee, Official Career Fair conducted by University of Moratuwa (2026)
- Member of the Organizing Committee, Project "Rest with Hope" conducted by Leo Club, UoM (2025)
- Active Member – FIT Moments (Faculty Media Body) | Leo Club of UoM | Legal Youth Community Sri Lanka

LANGUAGES

- English: Professional Working Proficiency (Oral & Written) | Sinhala: Native Proficiency

REFERENCES

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